

LOW MOLECULAR WEIGHT NBR: PATENT LANDSCAPE AND SYNTHESIS REPORT

1. ABSTRACT

This report reviews patent evidence for low molecular weight nitrile butadiene rubber (NBR), with emphasis on routes that reduce Mooney viscosity and narrow molecular weight distribution while preserving the native NBR microstructure. The primary patent in the manifest, EP1720919B1, discloses a post-polymerization depolymerization strategy rather than a direct low-MW polymerization route: nitrile rubber is reacted with a ruthenium- or osmium-based indenylidene metathesis catalyst, optionally in the presence of a co-olefin, to cleave unsaturation sites along the polymer backbone and generate lower molecular weight fragments. The process may be followed by hydrogenation to give optionally hydrogenated nitrile rubber with improved processability. The patent positions this approach against conventional mastication and strong-acid degradation, which can alter microstructure and broaden molecular weight distribution.

The disclosed target window is technically significant: the resulting hydrogenated nitrile rubber can have Mw of 30,000 to 250,000, Mooney viscosity (ML 1+4 at 100°C) of 3 to 50, and molecular weight distribution below 2.5. The evidence highlights the use of homogeneous metathesis catalysts based on Ru/Os indenylidene complexes, with ligand flexibility spanning phosphines, phosphites, amines, ethers, and related donor classes, and a non-coordinating anion. The process is framed as a controlled alternative to older depolymerization methods using Mo/W systems or $WCl_6/SnMe_4$, which were described as non-controlled and aimed at low molecular fragments rather than processable low-MW NBR.

Overall, the patent landscape in this manifest is narrow but technically focused: low-MW NBR is achieved by selective chain scission through olefin metathesis of unsaturated elastomer backbones, followed optionally by hydrogenation. The key formulation/process trend is preservation of microstructure while reducing viscosity and polydispersity, suggesting that catalyst choice, co-olefin use, and hydrogenation sequence are the main levers for tuning product performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP1720919B1
- Patent Title: Process for the preparation of low molecular weight nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a post-polymerization depolymerization process for nitrile rubber using homogeneous olefin metathesis. NBR is reacted with at least one ruthenium- or osmium-based indenylidene complex, optionally with a co-olefin, to cleave unsaturation sites and reduce molecular weight; the resulting product may then be hydrogenated by standard methods. - Catalyst metal: Ru or Os — Claim 1 / Abstract - Catalyst class: indenylidene complex — Abstract / Claim 1 - Co-olefin: optionally present — Claim 1 - Hydrogenation step: optional, after metathesis — Abstract / Claim 1 - Molecular weight (Mw): 30,000 to 250,000 — Description - Mooney viscosity (ML 1+4 @ 100°C): 3 to 50 — Description - Molecular weight distribution (MWD): less than 2.5 — Description - Mooney viscosity of commercially available NBR/HNBR: 55 to 105 — Background - Molecular weight of commercially available NBR/HNBR: 200,000 to 500,000 g/mol — Background - Polydispersity of commercially available HNBR: greater than 3.0 — Background - Residual double bond (RDB) content: 1 to 18% — Background - Mooney Increase Ratio (MIR): around 2 — Background - Mooney viscosity of Therban® VP KA 8837: 55 ML 1+4 @ 100°C — Background - RDB of Therban® VP KA 8837: 18% — Background - Conjugated diene chain length: C4-C6 preferred — Description - Unsaturated nitrile chain length: C3-C5 preferred — Description - Conjugated diene examples: butadiene, isoprene, piperylene, 2,3-dimethyl butadiene — Description - Unsaturated nitrile examples: acrylonitrile, methacrylonitrile — Description

Relevant Experimental Evidence - Example highlights: No numbered examples were detected in the provided evidence. - Example highlights: The disclosure demonstrates that metathesis can reduce NBR molecular weight while preserving microstructure. - Example highlights: The patent contrasts the process with mastication and strong-acid degradation, which broaden MWD and alter polymer structure. - Example highlights: The patent indicates that optional hydrogenation can be applied after depolymerization to obtain low-MW HNBR.

Technical Relevance This patent is directly relevant because it provides a controlled route to low molecular weight NBR without changing the copolymer microstructure. It identifies a catalyst-based depolymerization strategy that can lower viscosity and narrow dispersity, which are central goals for processable low-MW NBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence supports a catalyst-mediated depolymerization route as the most clearly disclosed strategy for low molecular weight NBR in this manifest. EP1720919B1 recommends homogeneous Ru/Os indenylidene metathesis, optionally with co-olefin and followed by hydrogenation, to obtain NBR/HNBR with substantially reduced Mw, lower Mooney viscosity, and narrower MWD while retaining the original microstructure. Based on the disclosure, the most relevant synthesis parameters are homogeneous metathesis catalysis, optional co-olefin participation, and post-metathesis hydrogenation, with the target product window of Mw 30,000 to 250,000 and Mooney 3 to 50.

5. REFERENCES

- EP1720919B1 | Process for the preparation of low molecular weight nitrile rubber | Not disclosed | EP | 2011 | <https://patents.google.com/patent/EP1720919B1/en>